MILITARY SPECIFICATION

WINCH, DRUM: POWER OPERATED

1. SCOPE

1.1 This specification covers one type of integrally constructed, helicopter transportable, power operated marine and shoreline service two-drum winch.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-E-485	-	Enamel, Semi-gloss, Rust-inhibiting.
TT-P-664	-	Primer, Coating, Synthetic, Rust-inhibiting, Lacquer-resisting.
UU-T-81	-	Tags, Shipping And Stock.

MILITARY

MIL-C-104	-	Crate, Wood, Lumber And Plywood Sheathed, Nailed And Bolted.
MIL-P-116	-	Preservation, Methods Of.
MIL-V-173	-	Varnish, Moisture And Fungus Resistance (For Treatment Of Communications, Electronic And Associated Equipment).
MIL-P-514	-	Plate, Identification, Instruction And Marking, Blank.
MIL-T-704	-	Treatment And Painting Of Materiel.
MIL-L-2105	-	Lubricating Oil, Gear, Multi-purpose.
MIL-E-16298	-	Electric Machines Having Rotating Parts And Associated Repair Parts, Packaging Of.
MIL-P-46093	-	Primer Coating, Synthetic, (For Brake Drums).

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STANDARDS

MILITARY

MIL-STD-129 - Marking For Shipment And Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association, Inc., Traffic Department, 1616 P Street, N. W., Washington, D. C. 20036.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION

IC-1 Industrial Control

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, N. Y., 10017.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

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3. REQUIREMENTS.

3.1 First article. The first article is defined as a production model which will be identical to the product which the supplier will subsequently produce in fulfillment of the contract. As soon as practicable after award of the contract or order and prior to submission of any complete units for final acceptance, the contractor shall furnish a first article for test under the direction and supervision of the Government inspector to determine conformance to this specification. Examination and tests shall be those specified herein. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to supply equipment conforming to this specification. Any changes or deviations of production units from the first article shall be subject to the approval of the contracting officer.

3.2 Parts and materials. Materials shall be as specified herein. Parts and materials not definitely specified shall be the best quality, suitable for the purpose, regularly used by the manufacturer or his suppliers in the product he sells on the commercial market and free from all defects and imperfections that might affect the serviceability and appearance of the finished product.

3.2.1 Fungus resistance. Electrical connections, including terminal and circuit connections, components, and circuit elements shall be coated with varnish conforming to MIL-V-173 except that:

(a) Components and elements inherently inert to fungi or hermetically sealed need not be treated.

(b) Components and elements whose operation will be adversely affected by the application of varnish shall not be treated.

(c) One percent of copper 8-quinolinonate (by weight based on the nonvolatile content of the varnish) may, at the supplier's option, be substituted for the salicylanilide specified in MIL-V-173, provided that the varnish conforms to MIL-V-173 in all other respects.

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3.3 Design and construction. The winch shall be an electric motor driven, two-drum type for use with 5/8- to 3/4-inch nominal size wire rope and shall be complete with built-on control stand and controls for brakes, clutches, electric motor, transmission and pawls. Each winch shall be furnished with a double fairlead and snatch blocks which may be separated from the winch. The winch, except for the accessories, shall be integrally assembled for maximum compactness and minimum weight. All rotating shafts and drums shall turn in antifriction bearings. The drums shall be positioned to allow for rope clearance. The forward drum shaft shall have a winch head installed on the side opposite the control station. The winch shall meet the requirements specified herein using 5/8-inch diameter wire rope. Wire rope shall not be furnished.

3.3.1 Foundation framing. The foundation and frame shall be constructed of rugged steel shapes, heavily reinforced at crossmembers and welded as a one-piece rigid base. The frame shall incorporate a skid base with reinforced holes to allow anchoring.

3.3.2 Control station. A control station shall be furnished as an integral part of the winch. Hand operated levers shall be provided to control clutches, drum holding devices, and to change transmission speeds. Brake pedals shall be provided with a latch and ratchet or other means to hold their position. Clutch controls shall be provided with a positive lock in the set and released positions.

3.3.3 Drums. The drums shall be of one-piece cast semisteel, cast steel or alloy steel construction. Drums shall have heavily reinforced barrels and flanges. Drum flanges shall incorporate finished brake and clutch friction surfaces. The drum barrel diameter shall be not less than 12-1/2 inches and the drum barrel length not less than 16 inches. The drum flange diameter and contour shall be adequate to provide operation clearance and storage capacity for not less than 1,050 feet of 5/8-inch nominal diameter wire rope. A means shall be provided for the attachment of 5/8-inch and 3/4-inch diameter wire rope without kinking, and with anchorage to prevent slippage.

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3.3.4 Winch head. One winch head shall be furnished, key mounted on the forward drum shaft, and located on the opposite side of the winch from the control station. The winch head shall be cast semisteel, cast steel or alloy steel construction. The barrel size shall be not less than 8 inches diameter and 11 inches long. The flange diameter shall be not less than 9-1/2 inches.

3.3.5 Drum clutches. Clutches shall be provided with a simple means for adjustment. Operation under maximum loading conditions for a continuous period of not less than 30 minutes shall not cause burning or damage to the lining.

3.3.6 Drum brakes. Brakes shall operate against a cast iron finished brake flange with a diameter not less than 28 inches. Brakes shall not slip when a static line pull of 20,000 pounds is applied tangent to the pitch circle of the second layer of 5/8-inch wire rope on the drum. There shall be no sign of eccentricity, nor change in clearance between brake band and brake ring, when the drum is rotated freely after 30 minutes of operation of the winch under full load conditions. A means to prevent backward rotation of the drums in case of power failure with the clutches engaged shall be furnished.

3.3.7 Drum holding. Ratchets or dog engaging wheels shall be cast integral with the drum, and shall have teeth or sockets of uniform contour to provide maximum surface contact with the pawl or dog. Control shall provide for full engagement or full release and shall have a snap action to prevent tooth skipping and accidental corner snagging. Pawls shall show no signs of distortion or breakage when locked in position, and loaded by the application of a line pull of not less than 20,000 pounds, with a rope lead tangent to the second layer.

3.3.8 Gears, pinions, sprockets and shafts. Gears, pinions and sprockets shall be of forged, heat treated, or cast alloy steel, of a structurally adequate alloy suitable for use near salt water. Rotating shafts shall be machined from alloy steel.

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3.3.9 Motor. The winch shall be powered with a 10 horsepower, 1,800 revolutions per minute, totally enclosed nonventilated squirrel cage motor.

3.3.9.1 Control box. The control box for the motor shall be mounted at the control station (see 3.3.2). The control box shall be a raintight enclosure conforming to NEMA IC-1, type 3R and shall include a winch ON-OFF switch, motor overload protection, and a receptacle for the power supply cable. The control box shall be designed and rated for use with the winches being furnished.

3.3.10 Accessories. The following accessories shall be provided with the winch.

3.3.10.1 Double fairlead. A double fairlead mounted in a rigid frame with a flat base shall be provided to allow for vertical and horizontal alignment of wire rope. The base shall have reinforced holes to allow anchoring the fairlead. The three fairlead sheaves shall each be 8 inches in diameter. The fairlead rollers shall be 4 inches in diameter.

3.3.10.2 Snatch blocks. Two snatch blocks, each with a 20-inch diameter sheave for 1-1/4-inch diameter wire rope, and one snatch block with a 10-inch diameter sheave for 5/8- to 3/4-inch diameter wire rope shall be furnished. Each snatch block shall be fitted with a shackle.

3.3.11 Lifting attachments. The winch shall be provided with permanently affixed lifting attachments. The attachments shall enable the winch to be lifted in its normal operating position. Each attachment shall withstand, without damage to any part of the winch, 2-1/2 times the resultant vectorial force imposed when the winch is suspended by the lifting attachments. Attachments for multiple slings shall be located so that slings will converge at not more than 19 feet above the lowest extremity of the winch, and not less than 1-inch clearance will be maintained between the slings and the winch. The inside diameter of each attachment eye shall be not less than 3 inches. Where required, suitable reinforcements shall be used to meet the requirements specified herein. Lifting attachments may also be used as tiedown attachments when such attachments conform to 3.3.12.

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3.3.12 Tiedown attachments. The winch shall be provided with suitable, permanently attached devices to permit tiedown of the winch to the floor or deck of the transportation medium. The tiedown devices shall be of such design, location, and number as to enable the winch to be anchored to prevent shifting or movement in any direction. The devices shall be designed to withstand thrusts of the following magnitude:

forward thrust 8 G

rearward thrust 8 G

upward thrust 2-1/4 G

sideward thrust 1-1/2 G

NOTE: G = cargo weight of unit

Tiedown devices shall be located so as to permit easy attachment of cables or turnbuckles.

3.3.13 Lubrication. Lubrication means shall be provided for all moving parts requiring lubrication. Pressure grease fittings shall be of the hydraulic type with surface check ball, and when applied to sealed bearings shall have pressure release to protect the seals.

3.3.14 Steel fabrication. Steel used in the fabrication of the winch shall be free from kinks and sharp bends. The straightening of material shall be done by methods which will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. All bends of a major character shall be made with controlled means in order to insure uniformity of size and shape.

3.3.15 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided in accordance with good commercial practice, and all bolts, nuts, and screws shall be tight.

3.3.16 Welding. Welding shall be in accordance with a nationally recognized welding code. The surface of parts to be welded shall be free from rust, scale, paint, grease, or other foreign matter. Spot, tack, or intermittent welds for strength shall not be permitted. Weld penetration

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shall be such as to provide transference of maximum design stress through the base metal juncture. Fillet welds shall be provided when necessary to reduce stress concentration. Before assigning any welder to manual welding work, the manufacturer shall certify to the contracting officer that the welder has passed the qualification tests as prescribed by a nationally recognized welding code, for the type of welding operation to be performed.

3.3.17 Castings. Castings shall be sound and free from patching, misplaced coring, warping, or other defects which might render the casting unsound for use.

3.3.18 Safety. The winch shall be designed to prevent conditions which may be hazardous to personnel or deleterious to equipment.

3.4 Maintainability. The design of the winch shall provide for maximum accessibility for maintenance and service in the field.

3.5 Performance. The winch shall be designed for continuous all weather use and shall not be damaged by occasional salt water splash. The winch shall be capable of maintaining a single drum line pull on either drum as shown in Table I, using 5/8-inch diameter cable.

Table I. Line Pull Requirements

Number of Layers on Drum	Line Pull Pounds (Min)	Speed Ft/Min (Max)
1	20,000	15
1	960	313
5	14,500	20.7
5	700	430
9	11,350	26.4
9	545	550

3.6 Dimensions. Overall dimensions of the winch shall not exceed 8 feet 6 inches long by 6 feet wide by 6 feet high.

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3.7 Weight. Total weight of the winch, including accessories, shall not exceed 8,000 pounds.

3.8 Treatment and painting. All exterior and interior surfaces of the winch, which are normally painted, shall be treated and painted in accordance with MIL-T-704, type B, except that the finish coat shall be in accordance with TT-E-485. Color shall match Marine Corps green, No. 23, (see 6.3).

3.9 Marking.

3.9.1 Nameplate. Each winch shall be equipped with a Marine Corps name and data plate. Plates will be furnished upon application to the Commandant of the Marine Corps, Headquarters U.S. Marine Corps (Code CSG), Washington, D. C. 20380. Data applicable to blank spaces 1, 3, 8, 9 and 10 on the plates shall be furnished by the Marine Corps. The manufacturer shall imprint data in all blank spaces provided and permanently affix the plates to the section in a conspicuous, protected location.

3.9.2 Instruction plates. The winch shall be equipped with instruction plates suitably located, describing location of lubrication fittings, lubricant required, frequency of lubrication, and any special or important procedures to be followed in operating and servicing the equipment. A data plate, visible to the operator, showing line pull and line speed for each transmission speed and for the first, fifth and ninth cable layers shall be furnished. Plates shall be of a material which will last and remain legible for the life of the equipment. Metal plates shall be securely affixed with brass screws or bolts of not less than 1/8-inch diameter.

3.9.3 Shipping data plates. Tiedown devices shall be indicated by a suitable information plate conforming to MIL-P-514, by stenciling, or other suitable marking of the winch. The marking shall clearly indicate that the device is intended for tiedown of the winch on the carrier when shipped. A diagram showing the lifting attachments and lifting slings shall be inscribed on a plate conforming to MIL-P-514, securely fastened to the winch. A silhouette of the winch showing the center of gravity shall be provided.

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3.10 Workmanship. Construction and finish shall be compatible with the best standard workmanship for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article (see 4.3).

(b) Acceptance (see 4.4).

(c) Preparation for delivery (see 4.5).

4.3 First article. The first article shall be subjected to the examination in 4.6 and the tests in 4.7.1 and 4.7.2.

4.4 Acceptance. Each winch shall be subjected to the examination in 4.6 and the test in 4.7.3.

4.5 Preparation for delivery. The preservation, packaging, packing and marking of the winches shall be inspected to verify conformance to the requirements of Section 5 of this specification.

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4.6 Examination. Each winch shall be examined for compliance with the requirements specified in Section 3 of this specification. Any redesign or modification of the supplier's standard product to comply with specified requirements, or any necessary redesign or modification following failure to meet specified requirements, shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement, or presence of one or more defects preventing or lessening maximum efficiency, shall constitute cause for rejection.

4.7 Tests. The winch shall be bolted to a suitable foundation and have sufficient 5/8-inch wire rope on each drum for test purposes. Failure of any test shall be cause for rejection.

4.7.1 Static line pull test. A static line pull of 20,000 pounds shall be applied to a wire rope lead tangent to the periphery of a full drum barrel second layer of rope. The load shall be held, using the drum pawls only for not less than 10 minutes. The load shall then be held by setting the brakes only. Repeat this test using the second drum. Any evidence of brake slippage, pawl and drum misalignment, any distortion, or failure of any component shall constitute failure of this test.

4.7.2 Rated line pull and speed test. Using 5/8-inch wire rope, the rated line pull and speed for each drum shall be determined. Inability to meet the requirements of Table I shall constitute failure of this test.

4.7.3 Operation test. The winch shall be operated to assure correct operation of all components. Failure of any component to operate correctly shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. The preservation and packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

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5.1.1.1 Disassembly and matchmarking. Disassembly shall be the minimum necessary to safeguard parts known to be subject to damage or loss and to accomplish reduction in cube. Bolts, nuts, screws, pins, and washers removed shall be reinstalled in one of the mating parts and secured to prevent their loss. When necessary to facilitate reassembly, parts removed and mating parts shall be matchmarked. Matchmarking shall be printed on tags conforming to UU-T-81, type A. The printed tags shall be waterproofed in accordance with MIL-STD-129.

5.1.1.2 Preservatives. Preservatives shall conform to the applicable specifications listed in, and shall be applied in accordance with, MIL-P-116.

5.1.1.3 Unpainted surfaces. Unpainted exterior metal surfaces including threaded surfaces and surfaces exposed by disassembly shall be coated with type P-1 preservative conforming to MIL-P-116.

5.1.1.4 Motors. Electric motors, component parts, and accessories shall be preserved in accordance with the level A requirements of MIL-E-16298.

5.1.1.5 Clutches. All surfaces of clutch assemblies enclosed within the housings, except composition facings, shall be sprayed or fogged with a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.6 Gears.

5.1.1.6.1 Exposed gears. Exposed gears shall be coated with type P-1 preservative conforming to MIL-P-116 or a thin film of primer conforming to TT-P-664 or MIL-P-46093.

5.1.1.6.2 Enclosed gears. Gear housings containing lubricant conforming to the applicable lubrication order, or to MIL-L-2105, shall be checked for proper operating level and the gears actuated to insure coating of all interior parts and surfaces. Gear housings not containing these lubricants shall be drained and refilled to the operating level with lubricant conforming to the applicable lubrication order and actuated as stated above.

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5.1.1.7 Drive chains.

5.1.1.7.1 Enclosed chains. Enclosed chains and chain housings shall be preserved as specified for enclosed gears.

5.1.1.7.2 Exposed chains. Exposed chains shall be coated with enough type P-3 preservative conforming to MIL-P-116 to assure penetration of the preservatives to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains and sprockets shall be coated with type P-1 preservative conforming to MIL-P-116.

5.1.1.8 Brakes. Brake drum facings shall be coated with a thin film of primer conforming to TT-P-664 or MIL-P-46093. Care shall be exercised to prevent the primer from coming in contact with the composition lining. Unpainted exposed surfaces of control mechanisms shall be coated with type P-1 preservative conforming to MIL-P-116.

5.1.2 Level C. The complete winches shall be preserved and packaged in accordance with the supplier's standard practice.

5.2 Packing. The packing shall be level A or C, as specified (see 6.2).

5.2.1 Level A. Each winch, together with packaged and unpackaged component parts, shall be packed in a crate conforming to MIL-C-104, type II, style a, class optional. The contents shall be anchored, blocked, and braced in accordance with the crate specification.

5.2.2 Level C. The complete winches shall be packed in a manner which will insure arrival at destination in satisfactory condition and which will be acceptable to the carrier at lowest rates. Packing shall comply with Uniform Freight Classification Rules or National Motor Freight Classification Rules.

5.3 Marking. Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The winch covered by this specification is intended for use in pulling rafts across a 300-foot wide water crossing in a 7-knot current.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following.

6.2.1 Procurement requirements.

(a) Title, number and date of this specification.

(b) Level of preservation, packaging and packing required (see 5.1 and 5.2).

6.2.2 Contract data requirements. Data conforming to Data Item Descriptions DI-M-4704, DI-R-4803 and DI-T-4902 will usually be required for delivery in connection with this specification. When so required, such data will be specified for delivery on a DD Form 1423 included in the contract.

6.3 Color. Color chips for Marine Corps green, No. 23, may be obtained from the Commandant of the Marine Corps (Code CSY), Headquarters U. S. Marine Corps, Washington, D. C. 20380.

6.4 First article. The first article tests shall be in accordance with Section 4.

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